

# Pathwise Perturbation Particle Filtering with Anticipated-Shock Quadrature and Piecewise Localization

Ed *(coauthor to be added)*

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## Abstract

We formalize a likelihood-evaluation strategy for nonlinear DSGE models based on *pathwise* (a.k.a. extended-path) perfect-foresight solves and first-order linearizations around those paths. At each filtering step we augment the model with an auxiliary “anticipated-shock” object—a finite horizon of future structural innovations—that is used to approximate conditional expectations in the equilibrium conditions. Conditional on this auxiliary object (and on a discrete *localization index* that encodes the relevant piece of the state space or the binding pattern of constraints), the model is linear-Gaussian: the one-step transition is Gaussian with parameters given by the pathwise Jacobians, and the measurement equation is assumed linear with Gaussian noise. Integrating out the auxiliary object yields a Gaussian-mixture transition kernel. This structure enables a Rao–Blackwellized particle filter that marginalizes the high-dimensional continuous state with a Kalman filter and samples only the low-dimensional auxiliary object and discrete localization index.

We derive the conditional linear-Gaussian representation from the stacked equilibrium system via the implicit function theorem, establish conditions for quadrature and localization refinement to converge to a well-defined mixture kernel, and provide stability-based bounds linking one-step kernel error to filtering and log-likelihood error. We also study a New Keynesian model with an effective lower bound (ELB) on the nominal rate and compare PPPF-based likelihood evaluation to OccBin and to PLC/COPF-style approximations.

**Keywords:** nonlinear DSGE estimation; particle filter; Rao–Blackwellization; extended path; stochastic extended path; dynamic perturbation; sigma points; occasionally binding constraints; effective lower bound.

**JEL:** C11, C15, C61, E32.

## 1 Introduction

Likelihood-based estimation of nonlinear DSGE models is difficult because equilibrium dynamics are nonlinear, typically available only implicitly via a system of equilibrium conditions with conditional expectations, and frequently feature kinks from occasionally binding constraints (OBCs) such as an effective lower bound (ELB). A direct likelihood approach nests the resulting nonlinear state-space system inside a particle filter, and for parameter inference typically nests that filter inside an MCMC/SMC scheme (Kantas et al., 2015; Andrieu et al., 2010). Two difficulties dominate: constructing a transition law that is accurate enough for likelihood-based inference, and controlling Monte Carlo variance of the likelihood estimator when the latent state is high-dimensional or the model has sharp nonlinearities.

A widely used strategy is to compute a global (or semi-global) approximation to the policy function (projection methods, value-function iteration, Smolyak/sparse grids, endogenous grid methods,

discrete regime grids, etc.) and then filter the resulting nonlinear state-space system. This can be accurate, but it can be computationally costly in medium-to-large DSGEs and becomes particularly challenging near kinks where standard bootstrap particle filters can suffer severe weight degeneracy.

This paper formalizes an alternative approximation strategy geared specifically toward likelihood evaluation. Instead of approximating the policy function globally, we approximate the *local* dynamics in the neighborhood of *paths* that are relevant under the data. At each filtering step we (i) solve for a finite-horizon equilibrium path using a perfect-foresight/extended-path system conditional on a low-dimensional auxiliary object that represents a finite horizon of “anticipated” future shocks, and then (ii) compute a first-order expansion around this path. This is in the spirit of the Extended Path (EP) method of [Fair and Taylor \(1983\)](#) and its stochastic generalizations ([Adjemian and Juillard, 2025](#)), and is closely related to Dynamic Perturbation ([Mennuni et al., 2025](#)), which computes first-order expansions along simulated equilibrium paths.

The central novelty is to embed this pathwise linearization inside a *filter*. Conditional on the auxiliary “anticipated-shock” object (and, when needed, on a discrete localization index encoding a regime/binding pattern or a partition cell in state space), the one-step transition is approximately linear-Gaussian, with parameters obtained from Jacobians of the stacked equilibrium system. Integrating out the auxiliary object by sigma points or other quadrature yields a Gaussian-mixture (Gaussian-sum) transition kernel ([Alsopach and Sorenson, 1972](#)). With linear-Gaussian measurement equations, this structure supports a Rao–Blackwellized particle filter (RBPF) that marginalizes a large continuous state block with a Kalman filter and samples only the low-dimensional auxiliary variables and discrete localization indices ([Doucet et al., 2000](#)).

The framework exposes a bias–variance–cost tradeoff. Finite-horizon solves, expectation approximations, and local linearization introduce approximation bias, while Rao–Blackwellization can substantially reduce Monte Carlo variance relative to a bootstrap particle filter ([Gordon et al., 1993](#)). Whether PPPF improves on global-solution-based likelihood evaluation depends on compute budgets, the effective dimension of the latent state, and how often the data drive the economy into regions where nonlinearities matter.

## 1.1 Contributions and organization

We derive a conditional linear-Gaussian approximation to DSGE transitions from stacked equilibrium conditions using the implicit function theorem, which clarifies the connection between EP/SEP, dynamic perturbation, and time-varying local linearizations along equilibrium paths. We then construct a Gaussian-mixture transition kernel by integrating over an anticipated-shock auxiliary variable using quadrature and explain when this quadrature can be interpreted as a probability mixture inside an RBPF. The paper also states an RBPF recursion tailored to this mixture structure, including mixture handling (branching versus sampling), general index proposals, and the role of localization indices for globalization and for regime selection in OBC models. Finally, we formulate an NK–ELB benchmark that permits direct comparison to OccBin and PLC/COPF-style approaches and separates approximation bias from Monte Carlo variability in likelihood evaluation.

The remainder of the paper is organized as follows. Section 3 introduces the nonlinear state-space model. Sections 4–6 derive the conditional linear-Gaussian approximation, its quadrature-induced mixture representation, and the role of discrete localization for globalization. Section 7 presents filtering and likelihood evaluation in detail. Sections 8–9 provide approximation-error decompositions and an explicit bias–variance–cost comparison to global-solution-based likelihood evaluation. Section 10 provides a minimal toy example that isolates the quadrature-as-mixture and mixture-handling logic. Section 11 specifies and studies the New Keynesian ELB benchmark. Appendices collect derivations and quadrature details.

## 2 Related literature

This section situates PPPF relative to work on nonlinear DSGE likelihood evaluation, OBC solutions, and mixture/Rao–Blackwellized filtering.

### 2.1 Nonlinear DSGE likelihood evaluation and particle filtering

Likelihood-based estimation of DSGE models with nonlinear or non-Gaussian state dynamics typically relies on sequential Monte Carlo methods, often embedded inside parameter-estimation schemes (Kantas et al., 2015; Andrieu et al., 2010). In macroeconomic applications, particle-filter likelihoods have been used for estimation and model comparison in nonlinear DSGEs; see, e.g., Fernandez-Villaverde and Rubio-Ramirez (2007) and the textbook treatment in Herbst and Schorfheide (2016). Variance reduction and robustness near sharp nonlinearities have motivated methods such as tempered particle filtering (Herbst and Schorfheide, 2019) and auxiliary/proposal-based particle filters (Pitt and Shephard, 1999).

### 2.2 Occasionally binding constraints and piecewise approximations

OBC models introduce non-differentiabilities that complicate both solution and filtering. Oc-cBin (Guerrieri and Iacoviello, 2015) and related finite-horizon regime-iteration methods produce piecewise-linear decision rules conditional on a binding pattern over a horizon, and have been widely used for ELB applications (Gust et al., 2017). Aruoba et al. (2021) develop piecewise-linear continuous (PLC) approximations for OBC DSGEs and a conditionally optimal particle filter (COPF) that exploits the resulting structure to reduce likelihood variance. PPPF is complementary: it uses pathwise linearizations and can incorporate an OBC binding pattern as a discrete localization index.

### 2.3 Extended-path, stochastic extended-path, and pathwise linearization

The extended-path method of Fair and Taylor (1983) solves a finite-horizon perfect-foresight system with a terminal condition, producing a time-varying local approximation to equilibrium dynamics that is often computationally attractive in medium-to-large models. Stochastic extended path (Adjemian and Juillard, 2025) improves on certainty equivalence by integrating over future shocks to better approximate conditional expectations. Dynamic perturbation (Mennuni et al., 2025) computes first-order expansions along simulated equilibrium paths and emphasizes accuracy along realized trajectories rather than at a fixed steady state. PPPF can be viewed as combining these ideas with a filtering recursion: build a pathwise linearization (conditional on a finite-dimensional shock-path proxy) and integrate it inside the likelihood evaluator.

### 2.4 Gaussian-sum filtering, sigma points, and Rao–Blackwellization

Gaussian-sum (mixture) representations of nonlinear filtering problems date back to Alspach and Sorenson (1972). Rao–Blackwellized particle filters (Doucet et al., 2000) exploit conditional linear-Gaussian structure to integrate out continuous components analytically via Kalman filtering, reducing Monte Carlo variance relative to a full bootstrap filter (Gordon et al., 1993). Sigma-point quadrature and related deterministic integration rules are widely used to approximate Gaussian expectations and underlie unscented filtering (Julier and Uhlmann, 2004). PPPF uses such quadrature not as a single-Gaussian moment-matching device, but to produce an explicit mixture kernel that can be handled either by branching or by sampling mixture indices within an RBPF.

| Symbol                                          | Meaning                                                                                |
|-------------------------------------------------|----------------------------------------------------------------------------------------|
| $x_t \in \mathbb{R}^{n_x}$                      | latent state (generic DSGE state vector used by the filter)                            |
| $y_t \in \mathbb{R}^{n_y}$                      | observed data                                                                          |
| $y_t = Mx_t + \eta_t$                           | measurement equation; $M$ is the measurement matrix                                    |
| $\eta_t \sim \mathcal{N}(0, R)$                 | measurement noise; $R$ is the measurement covariance                                   |
| $\varepsilon_{t+1} \sim \mathcal{N}(0, \Sigma)$ | realized one-step structural innovation (random)                                       |
| $H$                                             | EP/SEP horizon length (finite-horizon stacked system)                                  |
| $\omega_t \sim \mathcal{N}(0, \Omega)$          | auxiliary integration variable for expectations beyond $t + 1$                         |
| $\mu_\Omega(d\omega)$                           | Gaussian measure $\mathcal{N}(0, \Omega)(d\omega)$                                     |
| $k_t \in \{1, \dots, K\}$                       | quadrature node index; $\Pr(k_t = k) = w_k$ (if sampled as mixture)                    |
| $w_k$                                           | quadrature/mixture weights ( $w_k \geq 0$ if used as probabilities; $\sum_k w_k = 1$ ) |
| $s_t \in \{1, \dots, J\}$                       | localization index (partition cell or OBC regime/binding pattern)                      |
| $j_t = (s_t, k_t)$                              | combined discrete index used in the RBPF recursion                                     |
| $x_t^{\text{ref}}$                              | anchor/reference state for path solves and Jacobians                                   |

Table 1: Core notation.

### 3 Nonlinear state-space model

#### 3.1 Notation (random variables, indices, and measures)

Table 1 records the main objects and resolves common symbol collisions (notably:  $H$  is reserved for the EP horizon, and  $M$  denotes the measurement matrix). When we “integrate out” the auxiliary anticipated-shock object  $\omega$ , it is always with respect to the Gaussian reference measure  $\mu_\Omega(d\omega) = \mathcal{N}(0, \Omega)(d\omega)$ .

#### 3.2 Latent state and measurement

Let  $(x_t)_{t \geq 0}$  be an  $\mathbb{R}^{n_x}$ -valued latent state and  $(y_t)_{t \geq 1}$  observed data. We assume a linear measurement equation with Gaussian noise:

$$y_t = Mx_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, R), \quad (1)$$

where  $M$  and  $R$  may depend on parameters  $\theta$ .

#### 3.3 Equilibrium transition and Gaussian shocks

The state transition is induced by DSGE equilibrium conditions and structural innovations. For exposition we write it as a Markov mapping

$$x_{t+1} = F(x_t, \varepsilon_{t+1}; \theta), \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \Sigma), \quad (2)$$

even though in DSGEs  $F$  is usually not available in closed form.

Let  $K_\theta(x, \cdot)$  denote the true Markov kernel of  $x_{t+1} \mid x_t = x$  induced by (2).

## 4 From stacked equilibrium conditions to a conditional linear-Gaussian law

This section derives the object the filter will use: a Gaussian approximation to  $K_\theta(x_t, \cdot)$  conditional on an auxiliary object (anticipated shocks) and a discrete localization index.

## 4.1 A finite-horizon “stochastic perfect foresight” system

Fix a horizon  $H \geq 2$ . Let

$$\omega_t = (\varepsilon_{t+2}^a, \varepsilon_{t+3}^a, \dots, \varepsilon_{t+H-1}^a) \in \mathbb{R}^d, \quad d = \max\{0, H-2\}n_\varepsilon,$$

collect a sequence of *anticipated* shocks *beyond the one-step innovation* over the horizon (so for  $H = 2$ ,  $\omega_t$  is empty). We treat  $\omega_t$  as an auxiliary integration variable with the Gaussian law induced by the shock process:

$$\omega_t \sim \mathcal{N}(0, \Omega), \quad \Omega = \text{diag}(\Sigma, \dots, \Sigma),$$

where the block diagonal has  $H-2$  copies of  $\Sigma$  (and is vacuous when  $H = 2$ ). Conceptually,  $\omega_t$  is a numerical device used to approximate conditional expectations, not an additional structural shock, and it is *distinct* from the realized one-step innovation  $\varepsilon_{t+1}$  that moves the state from  $t$  to  $t+1$ .

Throughout, structural innovations  $\varepsilon_{t+1} \sim \mathcal{N}(0, \Sigma)$  are random. By contrast,  $\omega$  is an integration variable with respect to the Gaussian measure  $\mathcal{N}(0, \Omega)$  that we later approximate numerically by quadrature; it should not be interpreted as an additional shock realization on top of  $\varepsilon_{t+1}$ . Operationally, the stacked EP/SEP-style system is solved at a baseline  $\varepsilon_{t+1} = 0$ , while  $\omega_t$  perturbs the future-expectations terms over  $t+2, \dots, t+H-1$ .

Let  $X_{t:t+H-1} = (x_t, x_{t+1}, \dots, x_{t+H-1})$  and define a stacked residual mapping

$$\mathcal{R}(X_{t:t+H-1}; x_t^{\text{ref}}, \omega_t, \theta) = 0$$

encoding equilibrium conditions from  $t$  through  $t+H-2$  plus a terminal condition at  $t+H-1$  (e.g.  $x_{t+H-1} \approx x^*$ ). This is the finite-horizon stacked system solved in EP/SEP and related perfect-foresight solvers (Fair and Taylor, 1983; Adjemian and Juillard, 2025). We denote by

$$\bar{X}_{t:t+H-1} = \bar{X}(x_t^{\text{ref}}, \omega_t; \theta)$$

a solution of the stacked system.

## 4.2 Local perturbations around the path

In general the EP/SEP system may be solved (i) *per particle* at  $x_t$  or (ii) at a *reference/anchor state*  $x_t^{\text{ref}}$  (e.g. a weighted mean of particles) to avoid solving per particle. To keep the derivation aligned with implementable filters, we distinguish these cases explicitly.

Fix an anchor  $x_t^{\text{ref}}$  and let  $\bar{X}_{t:t+H-1} = \bar{X}(x_t^{\text{ref}}, \omega_t; \theta)$  denote the stacked solution path. Define deviations  $\tilde{x}_s = x_s - \bar{x}_s$  for  $s = t, \dots, t+H-1$ , so in particular  $\tilde{x}_t = x_t - x_t^{\text{ref}}$  because the initial condition in the stacked system is pinned at  $x_t^{\text{ref}}$ . Linearizing  $\mathcal{R}$  at  $\bar{X}$  yields a sparse block system linking  $\tilde{x}_{t:t+H-1}$  to *unexpected* innovations. To keep the derivation transparent, we focus on the one-step mapping from  $(x_t, \varepsilon_{t+1})$  to  $x_{t+1}$ . Let  $\varepsilon_{t+1}$  be the *unexpected* shock realization at  $t+1$  (mean zero). Then the linearized system implies

$$\tilde{x}_{t+1} = A_t(x_t^{\text{ref}}, \omega_t) \tilde{x}_t + B_t(x_t^{\text{ref}}, \omega_t) \varepsilon_{t+1}, \quad (3)$$

where  $A_t, B_t$  are obtained from the Jacobian of the stacked system (details in Appendix A).

Equation (3) yields a *conditional* linear-Gaussian transition:

$$x_{t+1} \mid (x_t, \omega_t, x_t^{\text{ref}}) \approx \mathcal{N}(m(x_t, \omega_t), Q(x_t, \omega_t)), \quad (4)$$

with

$$m(x_t, \omega_t) = \bar{x}_{t+1}(x_t^{\text{ref}}, \omega_t) + A_t(x_t^{\text{ref}}, \omega_t)(x_t - x_t^{\text{ref}}), \quad Q(x_t, \omega_t) = B_t(x_t^{\text{ref}}, \omega_t) \Sigma B_t(x_t^{\text{ref}}, \omega_t)^\top.$$

If one solves the stacked system per particle, one sets  $x_t^{\text{ref}} = x_t$  and the mean reduces to  $m(x_t, \omega_t) = \bar{x}_{t+1}(x_t, \omega_t)$ . If instead one uses a shared anchor  $x_t^{\text{ref}}$  (to amortize stacked solves across particles or to propagate a Gaussian belief with mean  $\mu_t$ ), the affine correction term  $A_t(x_t^{\text{ref}}, \omega_t)(x_t - x_t^{\text{ref}})$  transfers the pathwise linearization away from the anchor and constitutes an additional approximation.

*Remark 1* (Relation to dynamic perturbation and SEP). Dynamic perturbation (Mennumi et al., 2025) also computes first-order expansions along equilibrium paths. SEP (Adjemian and Juillard, 2025) integrates over future shocks to better approximate conditional expectations than certainty equivalence. The PPPF construction can be viewed as: (i) using a SEP-style draw/quadrature node  $\omega_t$  to build the reference path, (ii) computing a dynamic-perturbation-style linearization around that path, and (iii) integrating out  $\omega_t$  inside a filter.

## 5 Integrating out anticipated shocks

### 5.1 Gaussian-mixture transition kernel

Integrating out  $\omega_t$  yields an approximate transition kernel

$$\hat{K}_\theta(x_t, dx_{t+1}) = \int \mathcal{N}(dx_{t+1}; m(x_t, \omega), Q(x_t, \omega)) \phi(\omega; 0, \Omega) d\omega. \quad (5)$$

This is a Gaussian mixture (Gaussian-sum) transition (Alspach and Sorenson, 1972). In the per-particle version one sets  $x_t^{\text{ref}} = x_t$ , so (5) defines an approximate Markov kernel in  $x_t$  alone. In the reference-state shortcut, one holds  $x_t^{\text{ref}}$  fixed within the time- $t$  update (e.g. the weighted mean of particles) and interprets (4)–(5) as a controlled local approximation around that anchor. We approximate the integral by a  $K$ -node rule (sigma points, Gauss–Hermite, sparse grids, quasi-Monte Carlo, or Monte Carlo):

$$\hat{K}_{\theta, K}(x_t, dx_{t+1}) = \sum_{k=1}^K w_k \mathcal{N}\left(dx_{t+1}; m(x_t, \omega^{(k)}), Q(x_t, \omega^{(k)})\right), \quad \sum_{k=1}^K w_k = 1. \quad (6)$$

### 5.2 Sigma points as deterministic “anticipated shock” draws

For  $\omega \sim \mathcal{N}(0, \Omega) \in \mathbb{R}^d$ , the basic unscented transform uses  $2d + 1$  sigma points and associated weights that match mean and covariance and are exact for polynomials up to degree three (for symmetric choices) (Julier and Uhlmann, 2004). Sigma points are attractive when  $d$  is small (e.g. a small number of shocks and short horizon).

## 6 Discrete localization and “globalization” of the approximation

Sigma-point refinement controls the numerical integration error in (5), but does *not* by itself eliminate the local-linearization and horizon/terminal-condition errors. This section introduces a second approximation axis that can yield a genuine global refinement scheme.

## 6.1 Localization index: partition cells or regime/binding patterns

Let  $s_t \in \{1, \dots, J\}$  be a discrete localization index. Two cases are especially relevant. In the first,  $s_t$  indexes a partition of a compact set  $\mathcal{X} \subset \mathbb{R}^{n_x}$  into cells  $\{\mathcal{X}_j\}_{j=1}^J$  with anchors  $x^{(j)} \in \mathcal{X}_j$ . For each anchor (and each quadrature node  $\omega^{(k)}$ ) one computes  $(\bar{x}_{t+1}^{(j,k)}, A^{(j,k)}, B^{(j,k)})$  and defines a piecewise approximation by assigning each  $x_t$  to its cell and using the corresponding local parameters. When the continuous state is Rao–Blackwellized (stored as a Gaussian belief rather than as draws), this convention must be made explicit; a common choice is deterministic localization at an anchor/mean  $s_t = L(x_t^{\text{ref}})$  or  $s_t = L(\mu_{t-1})$ , while an alternative is probabilistic localization by cell mass under the Gaussian belief.

In the second,  $s_t$  encodes the binding/slack pattern of an occasionally binding constraint over a horizon (a regime path). OccBin (Guerrieri and Iacoviello, 2015) selects such a regime path by iterating on guesses until the constraint is satisfied, while filtering approaches can treat the regime as latent and integrate over it by branching or sampling. Aruoba et al. (Aruoba et al., 2021) construct globally piecewise-linear and continuous decision rules and derive a conditionally optimal particle filter (COPF) that exploits this structure. This regime conditioning is also the appropriate way to define pathwise Jacobians at kinks induced by max/min constraints: conditional on a regime the equilibrium system is smooth, Jacobians are well defined, and the filter can integrate over regimes via the discrete index.

## 6.2 A simple approximation theorem for localization refinement

To formalize “convergence to global” under partition refinement, we state a generic result. For simplicity, consider a setting where  $(m(x, \omega), Q(x, \omega))$  is smooth in  $x$  for each  $\omega$  (away from kinks). Let  $\hat{m}_J(x, \omega)$  and  $\hat{Q}_J(x, \omega)$  be piecewise-linear approximations built from anchors with mesh size

$$\Delta_J = \max_{j \leq J} \sup_{x \in \mathcal{X}_j} \|x - x^{(j)}\|.$$

**Assumption 1** (Smoothness in  $x$ ). *For each  $\omega$  in the support of the quadrature rule and all  $x \in \mathcal{X}$ , the maps  $x \mapsto m(x, \omega)$  and  $x \mapsto Q(x, \omega)$  are twice continuously differentiable with uniformly bounded second derivatives.*

**Proposition 1** (Localization refinement). *Under the smoothness assumption, there exists  $C < \infty$  such that*

$$\sup_{x \in \mathcal{X}} \|m(x, \omega) - \hat{m}_J(x, \omega)\| \leq C \Delta_J^2, \quad \sup_{x \in \mathcal{X}} \|B(x, \omega) - \hat{B}_J(x, \omega)\| \leq C \Delta_J^2,$$

for each fixed  $\omega$ , provided  $\hat{m}_J, \hat{B}_J$  are constructed by first-order Taylor (or equivalent local linear) approximations on each cell. Define  $\hat{Q}_J(x, \omega) = \hat{B}_J(x, \omega) \Sigma \hat{B}_J(x, \omega)^\top$ , which is positive semidefinite by construction. Then there exists  $C_Q < \infty$  such that

$$\sup_{x \in \mathcal{X}} \|Q(x, \omega) - \hat{Q}_J(x, \omega)\| \leq C_Q \Delta_J^2.$$

Hence if  $\Delta_J \rightarrow 0$  as  $J \rightarrow \infty$ , then the approximation errors vanish uniformly while preserving the covariance-matrix constraints needed by the Kalman recursion.

*Remark 2.* This is the standard “mesh  $\rightarrow 0$ ” globalization logic: piecewise linearization converges for smooth maps. For OBCs, the map is typically only piecewise smooth and the rate can degrade near the kink. In that case one refines the partition more aggressively near the kink set, or treats the kink via an explicit discrete regime as in OccBin and PLC/COPF.

## 7 Filtering and likelihood evaluation

### 7.1 Gaussian-sum representation

Under (6), the approximate transition can be written by introducing a discrete index  $k_t \in \{1, \dots, K\}$  with  $\Pr(k_t = k) = w_k$ :

$$x_{t+1} \mid (x_t, k_t = k) \sim \mathcal{N}(m_k(x_t), Q_k(x_t)).$$

With localization  $s_t$ , the index becomes  $(s_t, k_t)$ .

### 7.2 Rao–Blackwellized particle filter

Because the measurement is linear-Gaussian and the transition is conditionally Gaussian given  $(s_t, k_t)$ , we can marginalize the continuous state by Kalman filtering conditional on each particle’s discrete history. A Rao–Blackwellized particle filter (RBPF) samples  $(s_t, k_t)$  (or, in a continuous version, samples  $\omega_t$ ) and updates the conditional Gaussian state distribution analytically (Doucet et al., 2000).

### 7.3 PPPF-mixture-RBPF recursion

This subsection states the filtering recursion in a form suitable for likelihood evaluation. For ease of exposition we assume the mixture components are locally affine in  $x_t$ :

$$x_{t+1} \mid (x_t, s_t = s, k_t = k) \approx A_t^{(s,k)} x_t + a_t^{(s,k)} + \xi_{t+1}^{(s,k)}, \quad \xi_{t+1}^{(s,k)} \sim \mathcal{N}(0, Q_t^{(s,k)}),$$

and the measurement equation is linear-Gaussian as in (1). Each RBPF particle stores a Kalman-filter state (mean/covariance) for the continuous state conditional on its discrete history. Let  $\ell_t^{(i)}(j)$  denote the predictive likelihood of  $y_t$  under particle  $i$  and component  $j$ , obtained from the Kalman predict step under that component. The recursion can be summarized as follows.

1. Initialize each particle  $i$  with a Gaussian prior  $(\mu_0^{(i)}, P_0^{(i)})$  and particle weight  $W_0^{(i)} = 1/N$ .
2. For each  $t = 1, \dots, T$ :
  - (a) Choose an anchor state  $x_t^{\text{ref}}$  at which to build pathwise parameters. Two common choices are (i) per-particle anchors  $x_t^{\text{ref},(i)} = \mu_{t-1}^{(i)}$  (more accurate but more costly) and (ii) a shared anchor  $x_t^{\text{ref}} = \sum_i W_{t-1}^{(i)} \mu_{t-1}^{(i)}$  (computationally convenient but introducing anchor-reuse error).
  - (b) Build (or update) a mixture approximation to the one-step transition at the chosen anchor(s), producing component weights  $\tilde{w}_t^{(s,k)}$  and parameters  $(A_t^{(s,k)}, a_t^{(s,k)}, Q_t^{(s,k)})$ . In PPPF these parameters come from an EP/SEP-style stacked solve at  $(x_t^{\text{ref}}, \omega^{(k)}, s)$  followed by pathwise Jacobians, with  $Q = B\Sigma B^\top$ .
  - (c) For each particle  $i$ :
    - i. Sample a discrete index  $j_t^{(i)} = (s_t^{(i)}, k_t^{(i)})$  from a proposal  $q_t^{(i)}(\cdot)$ . A baseline choice is the prior  $q_t^{(i)}(j) = \tilde{w}_t^{(j)}$ ; an auxiliary proposal that conditions on  $y_t$  is  $q_t^{(i)}(j) \propto \tilde{w}_t^{(j)} \ell_t^{(i)}(j)$ .
    - ii. Kalman predict under the sampled component  $j_t^{(i)}$ :  $(\mu_{t|t-1}^{(i)}, P_{t|t-1}^{(i)})$ .
    - iii. Kalman update given  $y_t$ , obtaining  $(\mu_t^{(i)}, P_t^{(i)})$  and the predictive likelihood  $\ell_t^{(i)}(j_t^{(i)})$ .

- iv. Update the particle weight by multiplying by the predictive density and the proposal correction factor:

$$W_t^{(i)} \propto W_{t-1}^{(i)} \ell_t^{(i)}(j_t^{(i)}) \frac{\tilde{w}_t^{(j_t^{(i)})}}{q_t^{(i)}(j_t^{(i)})}.$$

- (d) Normalize weights, compute ESS, and resample if ESS falls below a threshold (e.g.  $0.5N$ ).
- (e) Accumulate the log-likelihood estimate by a stable log-sum-exp of the unnormalized particle weights.

The defining feature of the RBPF is that conditional on the discrete index  $j_t$  the continuous state remains linear-Gaussian and can be integrated analytically by a Kalman filter, reducing Monte Carlo variance relative to a bootstrap particle filter. Auxiliary index proposals that condition on  $y_t$  can further stabilize the recursion by targeting mixture components with high predictive likelihood (Pitt and Shephard, 1999). The dominant computational cost per period is governed by the number of evaluated components  $(s, k)$  and the cost of the associated stacked solves and Jacobian evaluations.

## 7.4 Mixture handling and quadrature choices

A central methodological point is *mixture handling*. Given a mixture transition  $\sum_k w_k \mathcal{N}(m_k, Q_k)$ , averaging the means  $m_k$  and covariances  $Q_k$  to obtain a single Gaussian is generally incorrect; it destroys tail behavior and Jensen effects. Instead, one must treat the mixture either by branching (expanding particles) or by sampling  $k$  and accounting for the mixture density in the proposal.

In the RBPF setting, branching corresponds to spawning children for each  $(s, k)$  and then resampling. Sampling corresponds to drawing  $(s, k)$  from their prior weights (or an auxiliary proposal) and updating particle weights accordingly.

When quadrature nodes are used as mixture components inside an RBPF, the weights must define a probability distribution. Sigma-point rules (e.g. the unscented transform) are often presented as deterministic quadrature and may assign negative weights under common parameterizations. Negative weights are acceptable for certain moment computations, but they are incompatible with sampling mixture indices. Accordingly, probability-mixture implementations require nonnegative-weight rules (e.g. positive-weight cubature, Gauss–Hermite with positive weights, or Monte Carlo with equal weights). Alternatively, the integral over  $\omega_t$  can be implemented by branching and aggregating signed contributions without interpreting quadrature weights as sampling probabilities.

For OBC models, a natural localization index is the binding pattern over the EP horizon (as in OccBin), which yields piecewise-linear dynamics conditional on the regime. More generally, localization can represent a partition of the state space, refined adaptively near kinks, so that first-order Taylor expansions are accurate within each cell. Section 6 formalizes this “globalization” logic.

## 8 Approximation theory: kernel, filter, and likelihood

### 8.1 Quadrature refinement converges to the mixture kernel

Let  $\hat{K}_\theta$  be the mixture kernel (5) and  $\hat{K}_{\theta,K}$  its  $K$ -node quadrature approximation (6).

**Assumption 2** (Quadrature consistency). *The quadrature rules  $(\omega^{(k)}, w_k)_{k=1}^K$  are consistent for*

the Gaussian measure  $\mathcal{N}(0, \Omega)$ :

$$\sum_{k=1}^K w_k h(\omega^{(k)}) \rightarrow \int h(\omega) \phi(\omega; 0, \Omega) d\omega \quad \text{for all bounded continuous } h.$$

**Proposition 2** (Weak convergence of  $\hat{K}_{\theta,K} \rightarrow \hat{K}_{\theta}$ ). *Assume  $(m(x, \omega), Q(x, \omega))$  is continuous in  $\omega$  for each fixed  $x$  and that dominated convergence applies to the Gaussian density integrand. Then for each fixed  $x$  and each bounded continuous test function  $\psi$ ,*

$$\int \psi(x') \hat{K}_{\theta,K}(x, dx') \rightarrow \int \psi(x') \hat{K}_{\theta}(x, dx').$$

*Remark 3.* This convergence holds to the mixture kernel  $\hat{K}_{\theta}$ , not to the true DSGE kernel  $K_{\theta}$ . Closing the remaining gap requires controlling the local-linearization error, horizon truncation/terminal-condition error, and expectation-approximation error in the underlying stochastic perfect-foresight step.

## 8.2 Kernel error decomposition

Write the total kernel error as

$$\begin{aligned} \|K_{\theta}(x, \cdot) - \hat{K}_{\theta,K,J}(x, \cdot)\| \leq & \underbrace{\|K_{\theta} - \tilde{K}_{\theta,H,p}\|}_{\text{horizon/expectation approximation}} + \underbrace{\|\tilde{K}_{\theta,H,p} - \hat{K}_{\theta}\|}_{\text{local linearization}} \\ & + \underbrace{\|\hat{K}_{\theta} - \hat{K}_{\theta,K}\|}_{\text{quadrature}} + \underbrace{\|\hat{K}_{\theta,K} - \hat{K}_{\theta,K,J}\|}_{\text{localization}}. \end{aligned} \quad (7)$$

where  $\tilde{K}_{\theta,H,p}$  denotes the kernel induced by the *exact* finite-horizon stochastic perfect-foresight solution with  $p$  future periods' expectations approximated by integration over  $\omega$  (no linearization), and  $\hat{K}_{\theta,K,J}$  denotes the computable PPPF kernel with  $K$  quadrature nodes and localization mesh index  $J$ . In the baseline construction in Section 4,  $\omega_t$  covers periods  $t+2, \dots, t+H-1$ , so  $p = H-2$ .

## 8.3 From kernel error to filter and likelihood error

Let  $\delta = \sup_{x \in \mathcal{X}} \|K_{\theta}(x, \cdot) - \hat{K}_{\theta,K,J}(x, \cdot)\|_{\text{TV}}$  be the worst-case one-step kernel error. Under standard mixing/forgetting conditions for hidden Markov models, errors in the transition kernel propagate to errors in filtering distributions and predictive likelihoods; see, e.g., [Del Moral \(2004\)](#); [Whiteley \(2013\)](#); [Kantas et al. \(2015\)](#).

**Assumption 3** (Uniform mixing and bounded likelihood). *There exists  $\epsilon \in (0, 1)$  and a probability measure  $\nu$  on  $\mathcal{X}$  such that for all  $x \in \mathcal{X}$  and measurable  $A$ ,*

$$K_{\theta}(x, A) \geq \epsilon \nu(A),$$

*and the measurement density satisfies  $0 < g_{\min} \leq g_{\theta}(y | x) \leq g_{\max} < \infty$  for all  $x \in \mathcal{X}$  and realized  $y_t$ .*

*Remark 4* (Strength of the global minorization condition). Assumption 3 is a standard sufficient condition used in stability theory, but it is often unrealistically strong for continuous-state DSGEs with near-deterministic transitions and informative measurements (where global minorization can fail). We include it only to make precise the qualitative message that *one-step transition-kernel*

errors propagate into filtering and likelihood errors under forgetting. In applied DSGE filtering one typically relies on local stability, bounded-moment conditions, and diagnostic checks rather than global minorization.

**Theorem 1** (Filter perturbation bound (standard form)). *Under the mixing and bounded-likelihood assumption, there exist constants  $C < \infty$  and  $\rho \in (0, 1)$  such that the filtering distributions satisfy*

$$\|\pi_{\theta,t} - \hat{\pi}_{\theta,t}\|_{\text{TV}} \leq C \sum_{s=1}^t \rho^{t-s} \delta, \quad t \geq 1.$$

**Corollary 1** (Log-likelihood perturbation bound (sketch)). *Under the same conditions, there exists  $C_\ell < \infty$  such that*

$$|\log p_\theta(y_{1:T}) - \log \hat{p}_\theta(y_{1:T})| \leq C_\ell T \delta$$

*up to geometric forgetting factors.*

## 9 When can PPPF be “better” than a global solution?

Let  $\ell_\theta$  denote the *true* log-likelihood and let  $\tilde{\ell}_\theta^{(A)}$  be a particle-based estimator using approximation strategy  $A \in \{\text{glob}, \text{pppf}\}$ . Under a compute budget  $C$ , the feasible particle count is roughly  $N_A \approx C/c_A$ , where  $c_A$  is cost per particle per period.

A standard decomposition is

$$\mathbb{E} \left[ (\tilde{\ell}_\theta^{(A)} - \ell_\theta)^2 \right] = \underbrace{(\text{Bias}_A)^2}_{\text{approximation bias}} + \underbrace{\frac{\sigma_A^2}{N_A}}_{\text{Monte Carlo variance}}.$$

Rao–Blackwellization reduces  $\sigma_{\text{pppf}}^2$  by integrating out (part of) the latent state analytically (Doucet et al., 2000). A sufficient “dominance” condition is therefore:

$$(\text{Bias}_{\text{pppf}})^2 + \sigma_{\text{pppf}}^2 \frac{c_{\text{pppf}}}{C} < (\text{Bias}_{\text{glob}})^2 + \sigma_{\text{glob}}^2 \frac{c_{\text{glob}}}{C}.$$

This inequality clarifies how relative approximation bias, Monte Carlo variance, and per-period computational cost jointly determine whether PPPF yields a lower mean squared error than a global-solution-based likelihood evaluator under a fixed compute budget.

## 10 A minimal toy example (no DSGE background)

The NK–ELB benchmark in Section 11 is economically meaningful but combines multiple approximation axes (finite horizon, expectation approximation, local linearization, and regime handling at a kink). This subsection records a minimal toy example whose “truth” can be computed to high accuracy and that isolates the *anticipated-shock quadrature* and *mixture-handling* components of PPPF.

Consider the following one-dimensional toy state-space model. Let  $x_t \in \mathbb{R}$  and  $y_t \in \mathbb{R}$  satisfy

$$y_t = x_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, r),$$

and define a transition that is conditionally linear-Gaussian given an auxiliary Gaussian variable  $\omega_t$ :

$$x_{t+1} \mid (x_t, \omega_t) = \rho x_t + b \tanh(\omega_t) + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1), \quad \omega_t \sim \mathcal{N}(0, \Omega).$$

Integrating out  $\omega_t$  yields a Gaussian mixture kernel of the form (5)–(6). This captures in the simplest possible way the PPPF logic: an auxiliary Gaussian object represents expectation-relevant nonlinear terms (here  $b \tanh(\omega)$ ), and quadrature produces an explicit mixture transition rather than a single Gaussian moment match.

This toy model isolates three issues. Certainty equivalence corresponds to setting  $\omega_t \equiv 0$ , which yields a single Gaussian transition with mean  $\rho x_t$ . Quadrature bias can be studied by comparing  $K = 1$  to increasing  $K$  in (6) under a nonnegative-weight rule when mixture indices are sampled. Finally, mixture handling can be studied by comparing (i) sampling or branching over mixture indices  $k_t$  to (ii) collapsing the mixture to a single Gaussian by moment matching, which is generally not likelihood preserving. Because the model is one-dimensional, a high-accuracy reference likelihood can be computed by fine-grid discretization or a large- $K$  Monte Carlo mixture approximation, which provides a transparent verification of the “quadrature-as-mixture” construction.

## 11 Benchmark: a New Keynesian model with an ELB

This section specifies a concrete benchmark that directly connects to the OccBin and PLC/COPF literatures.

### 11.1 Core equilibrium conditions

We consider a canonical three-equation New Keynesian model with an effective lower bound (ELB) on the nominal rate. Let  $\tilde{y}_t$  denote the output gap,  $\pi_t$  inflation, and  $i_t$  the nominal interest rate (deviations from steady state, appropriately scaled). A standard log-linearized NK system is

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \sigma^{-1} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n), \quad (8)$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t, \quad (9)$$

$$i_t^* = \phi_\pi \pi_t + \phi_{\tilde{y}} \tilde{y}_t + \nu_t, \quad (10)$$

$$i_t = \max\{\underline{i}, i_t^*\}. \quad (11)$$

Here  $r_t^n$  is a natural-rate (or preference) shock and  $\nu_t$  a monetary policy shock; both follow AR(1) laws with Gaussian innovations. Appendix C provides a brief microfoundation and discusses common variations (habit, indexation, wage rigidities).

The occasionally binding ELB constraint (11) introduces a kink and makes the decision rules nonlinear and piecewise smooth.

### 11.2 State-space representation and measurement

A minimal latent state can be

$$x_t^{\text{state}} = (\tilde{y}_t, \pi_t, i_t, r_t^n, \nu_t)^\top,$$

with observables

$$y_t = (\tilde{y}_t^{\text{obs}}, \pi_t^{\text{obs}}, i_t^{\text{obs}})^\top, \quad y_t = M x_t^{\text{state}} + \eta_t.$$

Because the ELB affects the transition, likelihood evaluation requires a nonlinear filter unless the solution imposes a structure that can be exploited.

### 11.3 Competing methods

The benchmark compares three likelihood-evaluation strategies that differ in how they approximate the transition kernel near the ELB kink and how they control Monte Carlo variance. First, an OccBin-style piecewise-linear extended-path solution (Guerrieri and Iacoviello, 2015) can be embedded in a bootstrap particle filter (or related inversion-filter variants when shocks are approximately identified). Second, the piecewise-linear continuous (PLC) approximation of Aruoba et al. (2021) delivers globally piecewise-linear decision rules and supports a conditionally optimal particle filter (COPF) that exploits the resulting structure; our benchmark includes a PLC-based baseline that filters only the exogenous state and uses a conditionally optimal proposal under a locally affine PLC measurement mapping. Third, PPPF constructs a Gaussian-mixture transition by integrating over a short-horizon anticipated-shock object  $\omega_t$  and, for OBC models, conditions on a discrete localization index  $s_t$  that selects a smooth binding or slack system. In ELB applications this regime conditioning also resolves the differentiability issue induced by the max operator, and motivates carrying a small mixture over regimes near the boundary rather than relying on hard regime selection.

### 11.4 Proposed evaluation metrics

Because particle filters yield stochastic likelihood approximations, the comparison emphasizes run-time per likelihood evaluation, Monte Carlo variability of repeated log-likelihood estimates on fixed data, ESS profiles (and resampling frequency), and impulse responses relative to the discretized global benchmark. The aim is to distinguish approximation bias (differences in mean likelihood/IRFs induced by different approximate kernels) from Monte Carlo variance (dispersion across repeated filter runs).

### 11.5 Experimental design

The empirical design combines an analytic verification exercise (Burnside recursion) for the expectation approximation component with a New Keynesian ELB benchmark. The benchmark experiment isolates the interaction between regime nonlinearity, mixture-based transitions, and likelihood evaluation.

In the NK-ELB experiment, the OccBin extended-path horizon is  $H = 20$ . We solve the discretized global benchmark (Tauchen grid) and simulate a dataset of length  $T = 250$  from that benchmark, adding small independent measurement noise on  $(\pi_t, \tilde{y}_t, i_t)$ . For likelihood evaluation we use  $N = 256$  particles and repeat each filter run  $N_{\text{rep}} = 30$  times with different random seeds to measure Monte Carlo variability.

To attribute PPPF likelihood bias in Table 3 to specific approximation steps, one can vary one component at a time. In particular, the ablations vary the quadrature resolution  $K$  (including  $K = 1$  certainty equivalence), the horizon and terminal condition  $(H, T_H)$ , the degree of anchor reuse, and the regime treatment at the ELB boundary. Reporting the mean log-likelihood gap to the discretized benchmark under these perturbations turns the bias-variance tradeoff into an explicit bias attribution exercise.

The discretized global benchmark used in this experiment induces a finite-state hidden Markov model (HMM) for the exogenous state, so its likelihood can be computed exactly by the standard forward recursion. Let  $g_t \in \{1, \dots, S\}$  index the Tauchen grid point for  $(r_t^n, \nu_t)$  and let  $P \in \mathbb{R}^{S \times S}$  be the resulting transition matrix. Solving the discretized global model delivers an expectation-consistent mapping from each grid point to observables  $h(g) = (\pi(g), \tilde{y}(g), i(g))^\top$ . With Gaussian

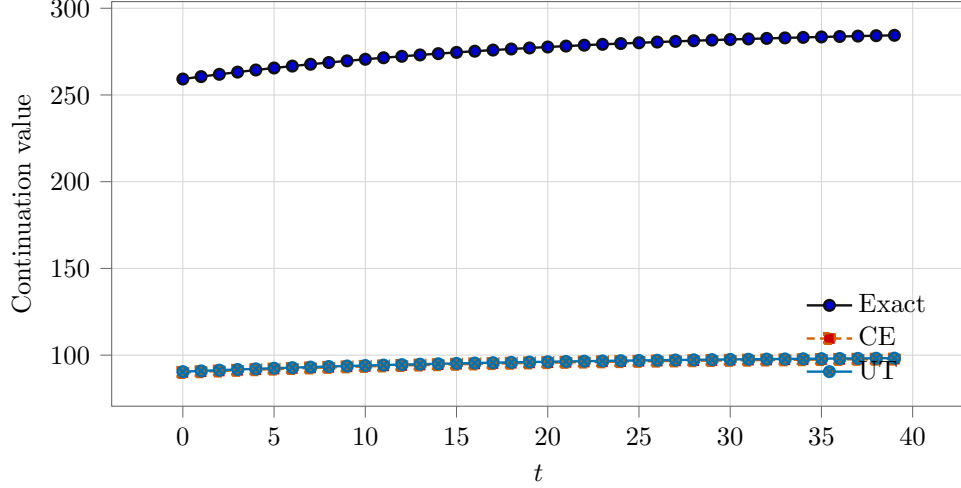


Figure 1: Burnside recursion verification exercise: exact benchmark versus certainty equivalence (CE) and unscented-transform (UT) quadrature.

measurement noise,  $y_t | g_t \sim \mathcal{N}(h(g_t), R)$ , the log-likelihood of the discretized benchmark can be computed exactly by the standard forward recursion:

$$\alpha_{t|t-1}(g') = \sum_{g=1}^S \alpha_{t-1}(g) P(g, g'), \quad \tilde{\alpha}_t(g) = \alpha_{t|t-1}(g) p(y_t | g), \quad z_t = \sum_{g=1}^S \tilde{\alpha}_t(g),$$

with  $\alpha_t(g) = \tilde{\alpha}_t(g)/z_t$  and  $\log p(y_{1:T}) = \sum_{t=1}^T \log z_t$ . This likelihood is exact for the *discretized benchmark* (grid + global policy functions), not for the underlying continuous-state DSGE.

## 11.6 Illustrative results

We begin with an analytic Jensen-style validation based on Burnside’s recursion, where an exact benchmark is available. Figure 1 reports impulse responses for the continuation value under the exact benchmark, a certainty-equivalence approximation, and a sigma-point (UT) one-step quadrature approximation. The UT approximation tracks curvature effects better than certainty equivalence, illustrating why explicit integration over a low-dimensional shock object can matter even when the underlying state is Gaussian.

As an additional verification exercise in a setting where all methods should agree, we validate the filtering layer on a one-dimensional linear-Gaussian state-space model with known Kalman-filter likelihood. In that case, an RBPF with a single mixture component ( $K = 1$ ) coincides with the Kalman filter and matches the exact log-likelihood to machine precision. A bootstrap particle filter, which replaces analytic Gaussian integration with Monte Carlo integration, approaches the same likelihood as the particle count increases.

To verify that the same logic applies in the New Keynesian environment, we rerun the NK model with the ELB effectively removed (setting  $\underline{i}$  far below any realized policy rate so the constraint never binds) and simulate data from the resulting linear-Gaussian state-space system driven by continuous AR(1) shocks. In this case an exact likelihood is available by Kalman filtering on the exogenous state  $(r_t^n, \nu_t)$  under the implied linear policy mapping; the “exact” baseline is constructed to match the same finite-horizon OccBin mapping used by the filters. Table 2 reports that PPPF

| Method                 | Mean loglik | SD loglik | Mean runtime (ms) | SD runtime (ms) | Mean ESS |
|------------------------|-------------|-----------|-------------------|-----------------|----------|
| Kalman (exact; no ELB) | 2345.0      | 0.0       | 0.0               | 0.0             | –        |
| OccBin-PF (no ELB)     | 2342.8      | 1.8       | 745.0             | 10.0            | 196.3    |
| PPPF-RBPF (no ELB)     | 2345.0      | 0.0       | 1226.6            | 8.9             | 2048.0   |

Table 2: NK no-ELB verification exercise: with the ELB inactive the model is linear-Gaussian, and PPPF with  $K = 1$  and  $\text{Var}(\omega) = 0$  coincides with the Kalman likelihood. Statistics are reported across  $N_{\text{rep}} = 10$  repeated filter runs on fixed data, with  $T = 250$  and  $N = 2048$  particles.

| Method            | Mean loglik | SD loglik | Mean time (ms) | SD time (ms) | Mean ESS |
|-------------------|-------------|-----------|----------------|--------------|----------|
| Benchmark (exact) | 2495.0      | 0.0       | 7.8            | 0.0          | –        |
| OccBin-PF         | 2199.4      | 8.9       | 101.0          | 1.8          | 27.6     |
| PPPF-RBPF         | 2105.2      | 1.9       | 285.9          | 6.5          | 179.3    |
| PLC-COPF          | 2405.6      | 1.3       | 18.1           | 0.1          | 160.8    |

Table 3: NK–ELB benchmark: repeated log-likelihood estimates on fixed data across  $N_{\text{rep}} = 30$  filter runs, with  $T = 250$  and  $N = 256$  particles. The benchmark likelihood is exact for the discretized model.

(degenerating to a single component by setting  $\text{Var}(\omega) = 0$ ) matches the Kalman likelihood to numerical precision, while OccBin-PF converges to the same likelihood up to Monte Carlo error.

We then turn to the NK–ELB benchmark. In what follows, OccBin-PF denotes OccBin coupled with a bootstrap particle filter, PPPF-RBPF denotes the PPPF mixture kernel coupled with a Rao–Blackwellized particle filter, and PLC-COPF denotes the PLC-based baseline with a conditionally optimal proposal under a locally affine PLC measurement mapping. Table 3 reports summary statistics from the benchmark comparison. We simulate data from the discretized global benchmark and compute its exact likelihood by the HMM forward algorithm described above. Relative to OccBin-PF, PPPF-RBPF delivers much higher ESS and substantially lower Monte Carlo variability of the likelihood estimator, consistent with the variance-reduction role of Rao–Blackwellization. We also report a PLC-based baseline obtained by interpolating the global (discretized) benchmark policies on the  $(r^n, \nu)$  grid and filtering only the exogenous state  $(r_t^n, \nu_t)$  using the conditionally optimal proposal associated with a local affine approximation of the PLC measurement mapping; this provides a concrete point of contact with the PLC/COPF perspective of [Aruoba et al. \(2021\)](#). The results separate Monte Carlo variance from approximation bias: Rao–Blackwellization and observation-informed proposals reduce Monte Carlo variability (high ESS and low dispersion across repeated runs), but approximation bias persists whenever the transition kernel used for likelihood evaluation differs materially from the data-generating benchmark (finite-horizon truncation, local linearization, low-dimensional anticipated-shock objects, or grid discretization).

To diagnose which approximation components drive PPPF bias in this calibration, Table 4 reports a one-factor-at-a-time ablation suite holding fixed the simulated dataset and (except where noted) the particle count and number of repeated filter runs. The reported gap is the PPPF log-likelihood minus the exact benchmark likelihood for the discretized model, so it isolates differences in the approximate transition kernel rather than Monte Carlo variability of the particle filter. The ablations illustrate that the ELB regime treatment matters sharply: fixing the time- $t+1$  binding decision across quadrature nodes produces substantial additional bias, while a two-regime mixture that averages bind/slack transitions using a Gaussian approximation to the binding probability partially restores likelihood accuracy at modest additional cost.

Figure 2 reports impulse responses near the ELB kink, comparing OccBin and PPPF to a

| Case                             | $H$ | $\omega$ | horizon | $K$ | Gap    | SD  | Mean ESS |
|----------------------------------|-----|----------|---------|-----|--------|-----|----------|
| Baseline                         | 20  |          | 1       | 5   | -389.9 | 1.8 | 180.0    |
| CE ( $K=1$ )                     | 20  |          | 1       | 1   | -277.6 | 0.0 | 256.0    |
| $\omega$ horizon 2               | 20  |          | 2       | 9   | -441.9 | 1.8 | 180.6    |
| $\omega$ horizon 3               | 20  |          | 3       | 13  | -498.5 | 2.2 | 172.7    |
| $H=10$                           | 10  |          | 1       | 5   | -522.3 | 1.3 | 187.2    |
| $H=40$                           | 40  |          | 1       | 5   | -395.2 | 3.7 | 181.0    |
| Fixed bind0                      | 20  |          | 1       | 5   | -615.6 | 9.0 | 179.2    |
| Bind0 mixture                    | 20  |          | 1       | 10  | -401.9 | 2.2 | 181.4    |
| Linear $i$                       | 20  |          | 1       | 5   | -358.4 | 0.9 | 179.2    |
| No aux proposal                  | 20  |          | 1       | 5   | -390.7 | 3.3 | 104.0    |
| Per-particle anchor <sup>†</sup> | 20  |          | 1       | 5   | -85.6  | 0.8 | 3.0      |

Table 4: NK–ELB PPPF ablations: mean log-likelihood gap to the exact discretized benchmark, standard deviation across repeated filter runs, and mean ESS. All rows use  $T = 250$ ,  $N = 256$ , and  $N_{\text{rep}} = 30$  unless marked with <sup>†</sup>, which indicates a reduced configuration ( $T = 50$ ,  $N = 4$ ,  $N_{\text{rep}} = 2$ ) used only to illustrate the effect of per-particle anchoring.

discretized global stochastic benchmark (expectation-consistent equilibrium on a Tauchen grid). Under this calibration, OccBin tracks the global benchmark more closely than PPPF in IRF RMSE, while PPPF exhibits substantially higher ESS (hence lower Monte Carlo variability) in likelihood evaluation. The lower PPPF likelihood relative to OccBin in Table 3 reflects approximation bias rather than particle-filter variance: PPPF is evaluating a different approximate kernel, and in this specification the anticipated-shock proxy and local linearization are not yet rich enough to match the discretized global benchmark.

## 12 Conclusion

PPPF replaces a global nonlinear approximation of DSGE policy functions with many local linearizations around stochastic perfect-foresight paths, integrating over an anticipated-shock object by sigma points/quadrature and optionally over regimes via a discrete localization index. With Gaussian shocks and linear measurement equations, this yields a Gaussian-mixture state-space model amenable to Rao–Blackwellized particle filtering. The paper derives the method from stacked equilibrium conditions, establishes convergence statements for quadrature and localization refinement, and links one-step kernel error to filtering and log-likelihood error using stability arguments. A New Keynesian ELB benchmark offers a natural comparison to the OccBin and PLC/COPF literatures.

The benchmark likelihood computed by the forward recursion is exact only for the discretized global model used to generate the data, not for the underlying continuous-state DSGE. Consequently, differences in likelihood across methods reflect both approximation error in the transition kernel and the discretization inherent in the benchmark. In addition, while the PLC-based baseline in Section 11 is designed to connect to the PLC/COPF perspective, a fully regime-exact COPF that integrates over PLC partition indices without local linearization remains a natural extension.

The NK–ELB results highlight that PPPF can deliver substantial variance reduction (high ESS, low Monte Carlo variability) while still exhibiting approximation bias when any of the four approximation axes is too crude. Reducing this bias requires sharpening the regime treatment at the ELB boundary (including regime mixtures near the kink), designing genuinely low-dimensional anticipated-shock objects so that positive-weight quadrature is feasible at moderate  $K$ , and implementing the ablation suite in Section 11 to attribute remaining discrepancies to  $(H, K)$ , anchor reuse, and regime handling.

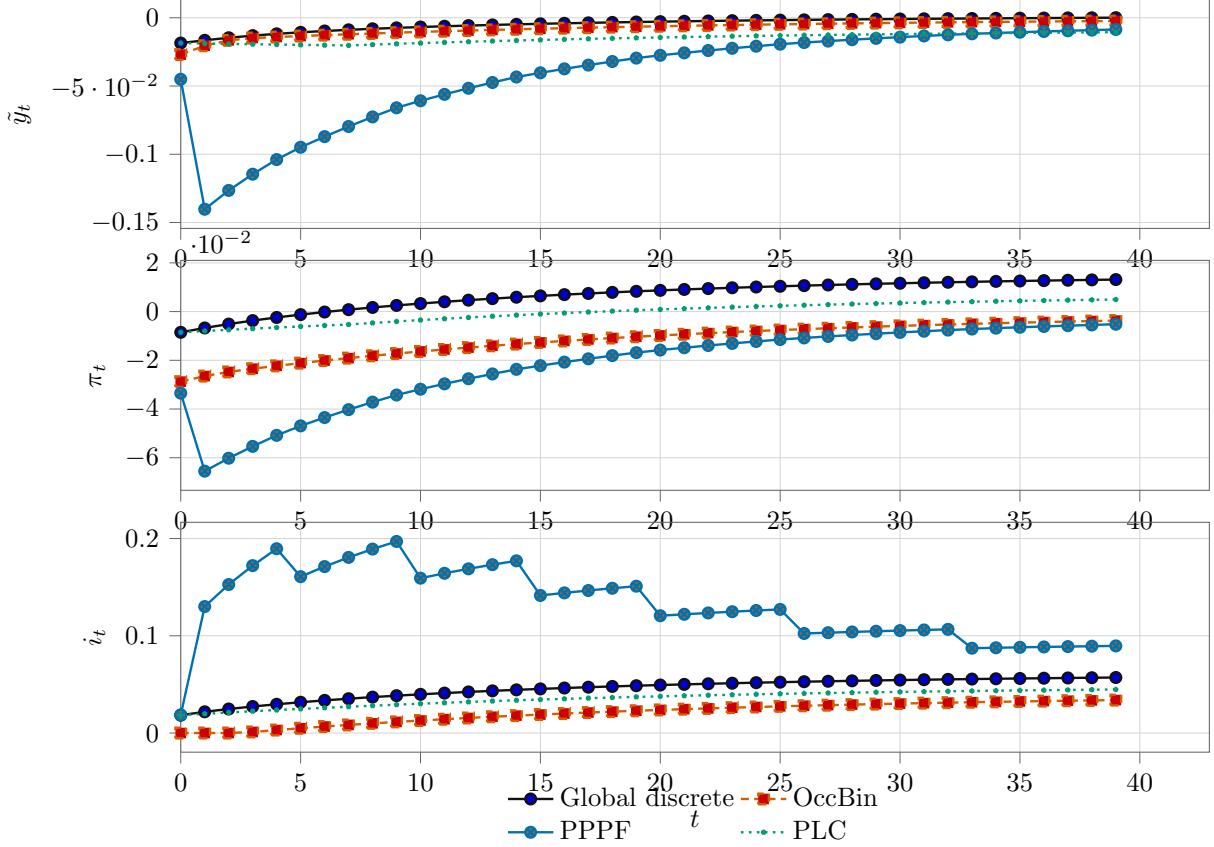


Figure 2: NK-ELB impulse responses: discretized global benchmark versus OccBin and PPPF-mixture approximations.

## A Deriving the conditional linear system via the implicit function theorem

This appendix provides a derivation of (3) from the stacked residual system. Let  $Y$  denote the stacked unknowns over the horizon excluding the pinned initial state:

$$Y = (x_{t+1}, x_{t+2}, \dots, x_{t+H-1}) \in \mathbb{R}^{(H-1)n_x}.$$

Write the stacked equilibrium conditions as

$$\mathcal{F}(Y; x_t, \omega_t, \theta) = 0.$$

Suppose  $\bar{Y}$  solves the system for given  $(x_t, \omega_t)$ . Assume the Jacobian with respect to  $Y$  is nonsingular:

$$D_Y \mathcal{F}(\bar{Y}; x_t, \omega_t, \theta) \text{ is invertible.}$$

Then by the implicit function theorem there exists a local mapping  $Y = G(x_t, \omega_t)$  with derivatives

$$D_x G = -(D_Y \mathcal{F})^{-1} D_x \mathcal{F}, \quad D_\omega G = -(D_Y \mathcal{F})^{-1} D_\omega \mathcal{F}.$$

To incorporate an *unexpected* shock  $\varepsilon_{t+1}$ , one augments the residual function to include the contemporaneous shock in the appropriate period- $t+1$  equations, and differentiates with respect to  $\varepsilon_{t+1}$ .

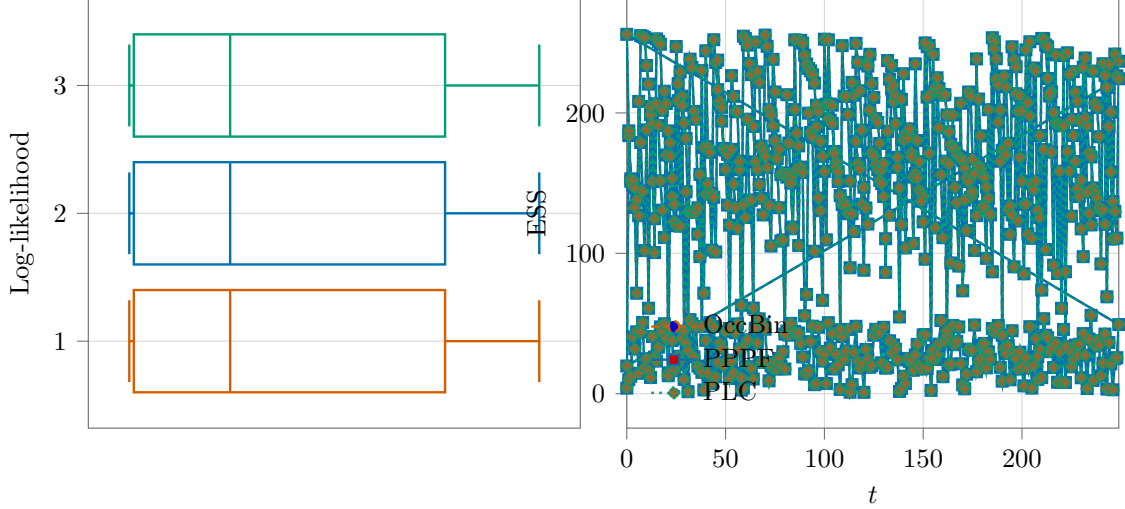


Figure 3: NK-ELB benchmark diagnostics: distribution of repeated log-likelihood estimates (boxplots) and ESS profile over time.

Because  $D_Y \mathcal{F}$  is sparse block-banded (typically block tridiagonal for one-step-ahead expectations), solving for these derivatives amounts to solving a sparse linear system, which is computationally attractive in large DSGEs.

Extracting the first block row of  $D_x G$  gives the mapping from  $x_t$  perturbations to  $x_{t+1}$  perturbations, yielding  $A_t$ . Similarly, the first block row of  $D_\varepsilon G$  gives  $B_t$ . Plugging into  $x_{t+1} = \bar{x}_{t+1} + \tilde{x}_{t+1}$  gives (3).

## B Unscented-transform sigma points

For  $\omega \sim \mathcal{N}(0, \Omega) \in \mathbb{R}^d$ , the basic sigma-point set uses  $2d + 1$  points:

$$\omega^{(0)} = 0, \quad \omega^{(i)} = + \left[ \sqrt{(d + \lambda)\Omega} \right]_{\cdot i}, \quad \omega^{(i+d)} = - \left[ \sqrt{(d + \lambda)\Omega} \right]_{\cdot i}, \quad i = 1, \dots, d,$$

with weights

$$w_0^m = \frac{\lambda}{d + \lambda}, \quad w_i^m = \frac{1}{2(d + \lambda)} \quad (i \geq 1),$$

and covariance weights  $w_0^c = w_0^m + (1 - \alpha^2 + \beta)$ ,  $w_i^c = w_i^m$ . Standard choices are  $\alpha \in (0, 1]$ ,  $\beta = 2$  (Gaussian prior),  $\kappa = 0$  (Julier and Uhlmann, 2004). If sigma points are used *as a probability mixture* (sampling a discrete index  $k_t$ ), the quadrature weights must satisfy  $w_k \geq 0$ . For the UT parameterization above, this requires  $\lambda \geq 0$  (hence  $w_0^m \geq 0$ ); the common “small- $\alpha$ ” settings used in unscented Kalman filtering typically violate this and should be treated as *signed quadrature* rather than a mixture distribution. When nonnegative weights are required, a simple positive-weight rule for  $\omega \sim \mathcal{N}(0, \Omega)$  uses  $2d$  points:

$$\omega^{(i)} = +\sqrt{d} \left[ \sqrt{\Omega} \right]_{\cdot i}, \quad \omega^{(i+d)} = -\sqrt{d} \left[ \sqrt{\Omega} \right]_{\cdot i}, \quad i = 1, \dots, d, \quad w_i = \frac{1}{2d}.$$

This yields a bona fide probability mixture (all weights equal and nonnegative) and is exact for polynomials up to degree three under a Gaussian reference measure.

## C Microfoundation and variants of the NK–ELB benchmark

Equations (8)–(9) arise from household Euler equations and Calvo price setting under standard assumptions; see [Woodford \(2003\)](#) for a textbook derivation. The ELB (11) is an occasionally binding constraint. In empirical work one often uses a shadow-rate specification or introduces additional wedges, but the essential computational feature is the kink introduced by the max operator.

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